Correcting Length Bias: Replicating and Adversarially Stress-Testing Pivoted Document Length Normalization

Intermediate Status Update · Information Retrieval · Group 6 · 25 October 2026

Divyansh Jain · Drishya Musaddi · Aditya Goyal · Himabh Gemini

GitHub: https://github.com/musaddi-drishya/Info_Ret_Project

1. Implementation Status & Architecture

The full from-scratch retrieval stack (Python + NumPy + Pandas + SQLite; no scikit-learn, Lucene or Elasticsearch in the core) is implemented and all five core experiments (E1–E5) plus two stretch experiments (E6 pivoted-unique, E7 BM25) run end-to-end on both public corpora in under 15 s. The architecture is unchanged from the proposal except for two additions: a TREC-style pooled judging module for the campus corpus and BM25/pivoted-unique scorers sharing the same index.

Component	Status	Notes
Preprocessing (tokenizer, stopwords, Porter stemmer)	Done	Stemmer written from scratch; spot-checked against reference outputs
SQLite storage + inverted index + .npz cache	Done	Dict postings frozen into CSR NumPy arrays
TF-IDF + cosine (E1), pivoted cosine (E2)	Done	ltc weighting; pivot = mean cosine norm
20NG / Reuters samples, 55 queries	Done	1,000 docs each, stratified by category × length quintile
Metrics, length-bias analysis, time/memory	Done	P/R/F1@5,10,20, MAP, NDCG@10, TVD, log-length gap
Verbosity (E3) and multi-topic (E4) attacks	Done	Fixed-length synthetic docs; relevance by construction
Slope sweep (E5), clean and attacked	Done	s = 0.0 … 1.0, step 0.1
Stretch: pivoted-unique (E6), BM25 + b-sweep (E7)	Done	Ahead of proposal schedule (planned week 6)
Custom campus corpus	Collected	100 BITS Pilani news/notice pages crawled; 0/205 pooled pairs double-judged
Failure-mode write-up, OCR-noise stretch, demo	Pending	See Section 4

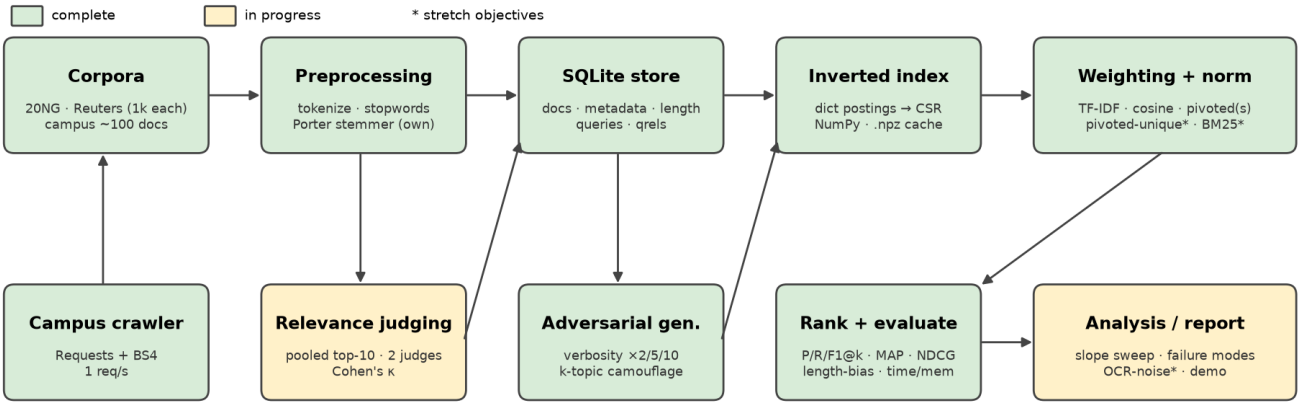


Figure 1. System architecture (green = complete, yellow = in progress).

2. IR Pipeline Details

Data. 20 Newsgroups: headers removed (they leak the label) except *Subject*; quoted replies and signatures stripped. Reuters-21578: single-topic documents from 20 topics. Each corpus is sampled to 1,000 documents, stratified by category and by length quintile within category; documents with fewer than 20 index terms are dropped. 30 (20NG) and 25 (Reuters) short topic queries were hand-written; relevance = category membership (proxy, as declared in the proposal). The campus corpus is crawled with Requests + BeautifulSoup at 1 request/s, de-duplicated by slug and content hash; because the portal posts far fewer than 100 items a week, the 100 pages span Jan 2025 – Oct 2026. 18 campus queries are drafted and a 205-pair judging pool (pooled top-10 of Cosine, Pivoted and BM25) is ready.

Preprocessing. Lower-casing, regex tokenisation (`[a-z][a-z0-9]+`), ~180-word stopword list, Porter stemming (own implementation, steps 1a–5b, memoised).

Indexing. Term → {doc: tf} dictionary postings are frozen into CSR arrays (`term_ptr, doc_ids, tfs`) so that each query term is one vectorised slice (term-at-a-time accumulation). Document length, unique-term count and mean tf are stored per document; the index is cached as compressed .npz. 20NG: 19,257 terms, 81,003 postings, 2.1 MB, built in 1.83 s; Reuters: 6,984 terms, 63,455 postings, 1.0 MB, 0.85 s.

Retrieval models. score(q,d) = Σ_t $w_q(t) \cdot w_d(t,d) / N_d$ with w = (1+ln tf)·ln(N/df). **Cosine**: $N_d = ||d||_2$. **Pivoted cosine**: $N_d = (1–s)\cdot p + s\cdot||d||_2$, p = mean $||d||_2$, default s = 0.7 (s = 1 recovers cosine). **Pivoted unique** (Lnu): $w_d = (1+\ln tf)/(1+\ln avg\text{-}tf_d)\cdot idf$, $N_d = (1–s)\cdot p + s\cdot|unique(d)|$, s = 0.2. **BM25**: $k_1 = 1.2$, b = 0.75. Slope sweeps reuse cached posting weights and only recompute N_d.

Evaluation. P@k, R@k, F1@k (k = 5, 10, 20), MAP over full rankings, NDCG@10, query latency and index memory. Length bias follows Singhal et al.: documents are split into 10 equal-frequency length bins and P(retrieved | bin) over the top-50 of every query is compared with P(relevant | bin). We summarise it as the total-variation distance (TVD) between the two distributions and the *log-length gap* = mean ln|d| of retrieved – of relevant (negative ⇒ short-document preference). **Attacks.** E3 replaces 50 random

documents by ×2/×5/×10 repetitions of themselves (no new information). E4 injects 20 synthetic 300-token documents per condition built from 1, 2, 3 or 4 categories (equal share each), plus a *verbose* control (one short document repeated to 300 tokens); each is relevant to all queries of its source categories.

3. Preliminary Evaluation Results

Corpus	Model	P@10	R@10	F1@10	P@20	MAP	NDCG@10	TVD	LenGap	ms/q
20NG	Cosine (baseline)	0.713	0.143	0.238	0.657	0.373	0.742	0.093	+0.171	0.08
20NG	Pivoted s=0.7	0.730	0.146	0.243	0.655	0.374	0.757	0.144	+0.292	0.06
20NG	Piv-unique*	0.740	0.148	0.247	0.652	0.371	0.756	0.171	+0.336	0.08
20NG	BM25*	0.733	0.147	0.244	0.645	0.374	0.754	0.139	+0.277	0.09
Reuters	Cosine (baseline)	0.932	0.184	0.307	0.858	0.714	0.946	0.050	+0.052	0.20
Reuters	Pivoted s=0.7	0.920	0.182	0.304	0.850	0.714	0.935	0.089	+0.133	0.15
Reuters	Piv-unique*	0.912	0.180	0.301	0.840	0.708	0.933	0.127	+0.197	0.15
Reuters	BM25*	0.952	0.188	0.314	0.882	0.725	0.965	0.106	+0.152	0.11

Table 1. Clean-collection effectiveness, length bias and latency (* stretch models; log-len gap > 0 = long-document preference).

RQ1 / RQ2 — length bias. Our cosine baseline does *not* reproduce the short-document preference reported on TREC: on both corpora it already retrieves long documents slightly more often than they are relevant (log-length gap +0.171 on 20NG, +0.052 on Reuters; Figure 2 left). Pivoting therefore moves the system further from the relevance curve (TVD 0.093 → 0.144 on 20NG) while MAP is essentially flat (0.3729 → 0.3735; best slope s = 0.7 / 0.8). The reason is the evaluation set-up: with category-proxy relevance and length-stratified sampling, P(relevant | length) is flat, whereas pivoting is designed for collections where it rises with length. This is the main motivation for the human-judged campus corpus (Section 4).

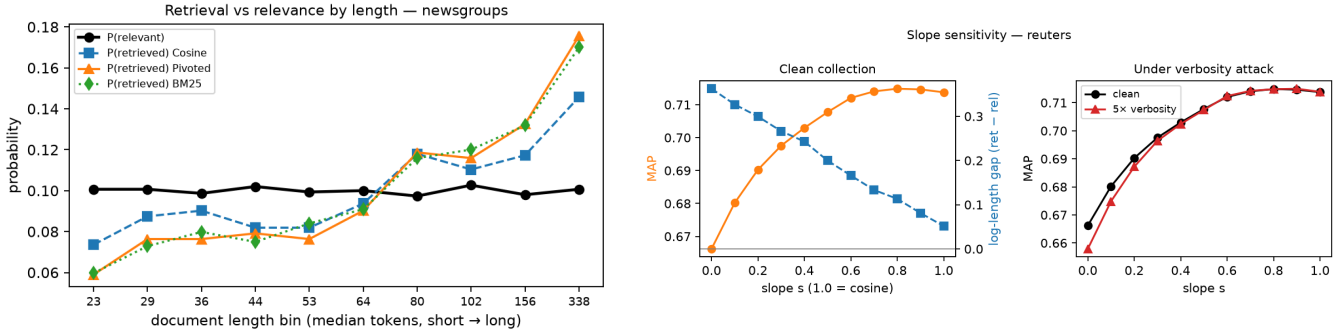


Figure 2. Left: retrieval vs relevance by length bin (20NG). Right: MAP and log-length gap vs pivot slope on Reuters, clean and under ×5 verbosity padding.

Model	P@10 (1×→10×)	R@10	F1@10	MAP	padded-doc median rank	intrusions@10
Cosine	0.932 → 0.924	0.184 → 0.183	0.307 → 0.305	0.7138 → 0.7136	34 → 38	0 → 0
Pivoted	0.920 → 0.920	0.182 → 0.182	0.304 → 0.304	0.7140 → 0.7140	36 → 30	1 → 2
PivUnique	0.912 → 0.912	0.180 → 0.180	0.301 → 0.301	0.7077 → 0.7068	45 → 45	3 → 2
BM25	0.952 → 0.944	0.188 → 0.187	0.314 → 0.312	0.7246 → 0.7178	36 → 28	2 → 4

Table 2. Verbosity attack on Reuters, before (1×) → after (10× padding of 50 documents). Intrusions = padded docs entering the top-10 of queries they are not relevant to.

RQ3 — verbosity. Repetition adds no information, so an ideal normaliser should leave rankings unchanged. Cosine slightly *demotes* padded documents, pivoted cosine and BM25 *promote* them, and pivoted-unique is nearly immune: repetition leaves its denominator (unique-term count) unchanged and dividing by 1+ln(avg-tf) largely cancels the inflated tf (exactly when tf equals the document average). Aggregate P/R/F1/MAP move by less than 0.01 because only 5% of the collection is attacked, but the effect grows sharply as the slope falls (Figure 2 right): at s = 0 the padded documents jump from median rank 44 to 14 and cause 14 top-10 intrusions, versus 1 at s = 0.7 and 0 at s = 1.

Median rank of synthetic doc (300 tokens)	Cosine	Pivoted s=0.7	Pivoted s=0.3	Piv-unique	BM25
verbose (1 doc repeated)	31	22	14	38	21
1-topic	19	14	7	11	12
2-topic	36	31	18	25	26
3-topic	41	35	27	31	34
4-topic	62	54	41	50	51

Table 3. Multi-topic camouflage on Reuters: median rank of injected documents for the queries they are relevant to (lower = better).

RQ4 — multi-topicality. At equal length, every model ranks genuinely multi-topic documents far below single-topic ones (cosine: median rank 19 → 62 from 1 to 4 topics): each topic gets only a quarter of the term mass, so length normalisation alone cannot tell *diluted* from *verbose*. Lower slopes help multi-topic documents, but they reward the verbose control by the same mechanism (pivoted s = 0.3: verbose 14 vs 2-topic 18). **RQ5 — slope.** MAP is flat for $s \in [0.6, 1.0]$ and drops steeply below 0.5, which is exactly where verbosity-padded documents begin to intrude: the slope that is safe on clean data is also the one that limits the attack.

4. Path to Final Submission (remaining ≈ 20–25%)

Work item	Owner	Target
Campus corpus: finalise 15–20 queries, double-judge pooled pairs, Cohen's κ, run E1–E5 on it	Drishya, Aditya	1–3 Nov
Held-out slope tuning (split queries) to remove the optimistic bias of E5; paired randomisation significance tests for Table 1	Divyansh	3–5 Nov
Failure-mode analysis: per-query case studies of intrusions and multi-topic misses; pivoted-unique under multi-topic camouflage	Himabh	3–7 Nov
Stretch: OCR-like noise (character substitutions) × length normalisation	Himabh, Aditya	7–10 Nov
Final report, slides, live CLI search demo comparing models side by side	All	10–14 Nov

Stretch objectives. Pivoted-unique normalisation (E6) and BM25 with a b-sweep (E7) are already implemented and included above, ahead of the planned week-6 slot. OCR-noise robustness has not started.

5. Contribution Matrix

Member	Components owned and completed to date	Status
Divyansh Jain	Inverted index (dict postings → CSR NumPy, .npz cache); scoring models: TF-IDF cosine, pivoted cosine, pivoted-unique, BM25; experiment runner and search CLI	Complete
Drishya Musaddi	Corpus pipeline: 20NG / Reuters parsing, header/quote stripping, stratified sampling; SQLite schema; campus crawler; judging pool	Complete; campus judging in progress
Aditya Goyal	Preprocessing (tokenizer, stopwords, from-scratch Porter stemmer); evaluation metrics; retrieval-vs-relevance length-bias analysis; figures	Complete
Himabh Gemini	Adversarial generator (verbosity padding, multi-topic camouflage, verbose control); attack experiments E3, E4 and the stretch comparisons E6/E7; live attack demo; this report	Complete; failure analysis ongoing